

On-Chain (EVM) — immutable, auditable

Reserve balances

Loan requests & approval /
rejection events

Repayment transactions

Role bindings (RBAC)

Borrowing-limit
enforcement

Off-Chain (PostgreSQL) — private, queryable

User profiles & KYC
documents

Chat / session history

ML risk features & scores

Security & agent action logs

Market data cache

Event listener
borrowing-limit sync
protocol

```
graph TD; subgraph OnChain [On-Chain (EVM) — immutable, auditable]; R1[Reserve balances]; R2[Loan requests & approval / rejection events]; R3[Repayment transactions]; R4[Role bindings (RBAC)]; R5[Borrowing-limit enforcement]; end; subgraph OffChain [Off-Chain (PostgreSQL) — private, queryable]; R6[User profiles & KYC documents]; R7[Chat / session history]; R8[ML risk features & scores]; R9[Security & agent action logs]; R10[Market data cache]; end; OnChain --> EL[Event listener borrowing-limit sync protocol]; OffChain --> EL;
```

The diagram illustrates a data architecture with two main data sources at the top and a central processing unit at the bottom. The left source, 'On-Chain (EVM) — immutable, auditable', contains five light blue boxes: 'Reserve balances', 'Loan requests & approval / rejection events', 'Repayment transactions', 'Role bindings (RBAC)', and 'Borrowing-limit enforcement'. The right source, 'Off-Chain (PostgreSQL) — private, queryable', contains five light blue boxes: 'User profiles & KYC documents', 'Chat / session history', 'ML risk features & scores', 'Security & agent action logs', and 'Market data cache'. Arrows from the bottom of both source containers point towards a central light gray box at the bottom labeled 'Event listener borrowing-limit sync protocol'.